

What Ancient Texts Remember and Inscriptions Forget

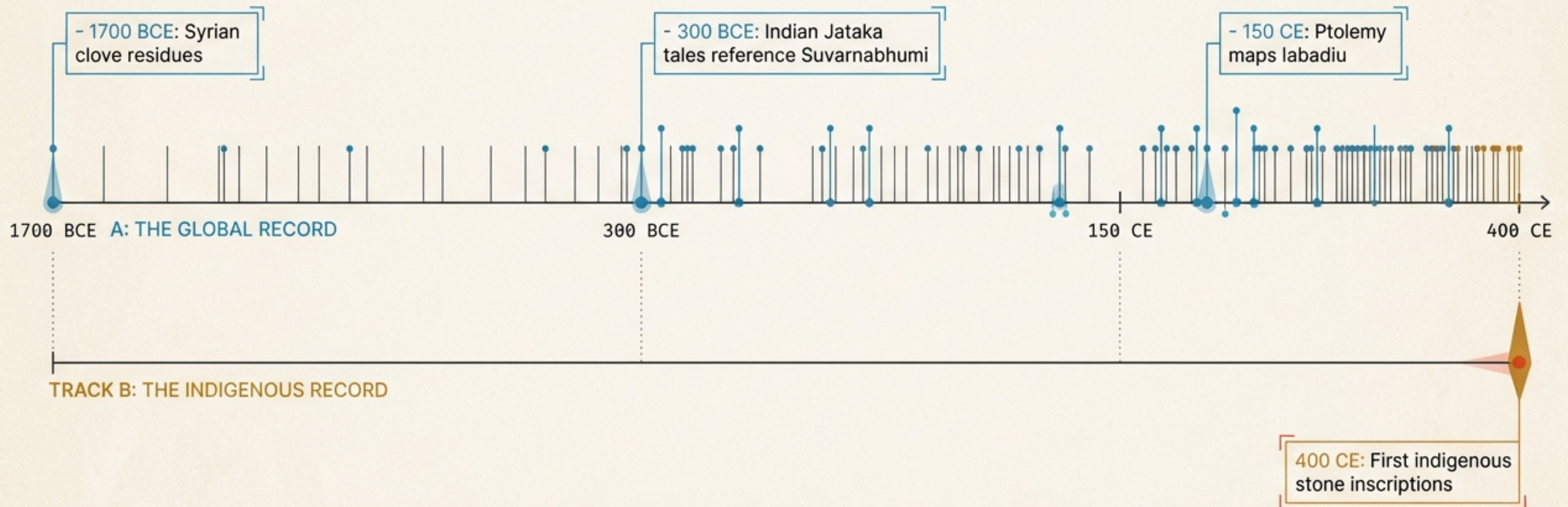
Transformer-Based Evidence for Volcanic Taphonomic Bias in Indonesian Textual Records

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The Paradox of Archaeological Darkness

Indonesia has been inhabited for over a million years and traded globally for millennia. Yet, the indigenous inscripational record in stone begins abruptly around 400 CE. If complex societies existed here, why are they invisible in the earth?



The Three Suspects: The VOLCARCH Framework

Physical decay biases the archaeological record.
But what if the texts themselves are biased?
The stone doesn't just weather; it filters.

3. Textual Genre Taphonomy

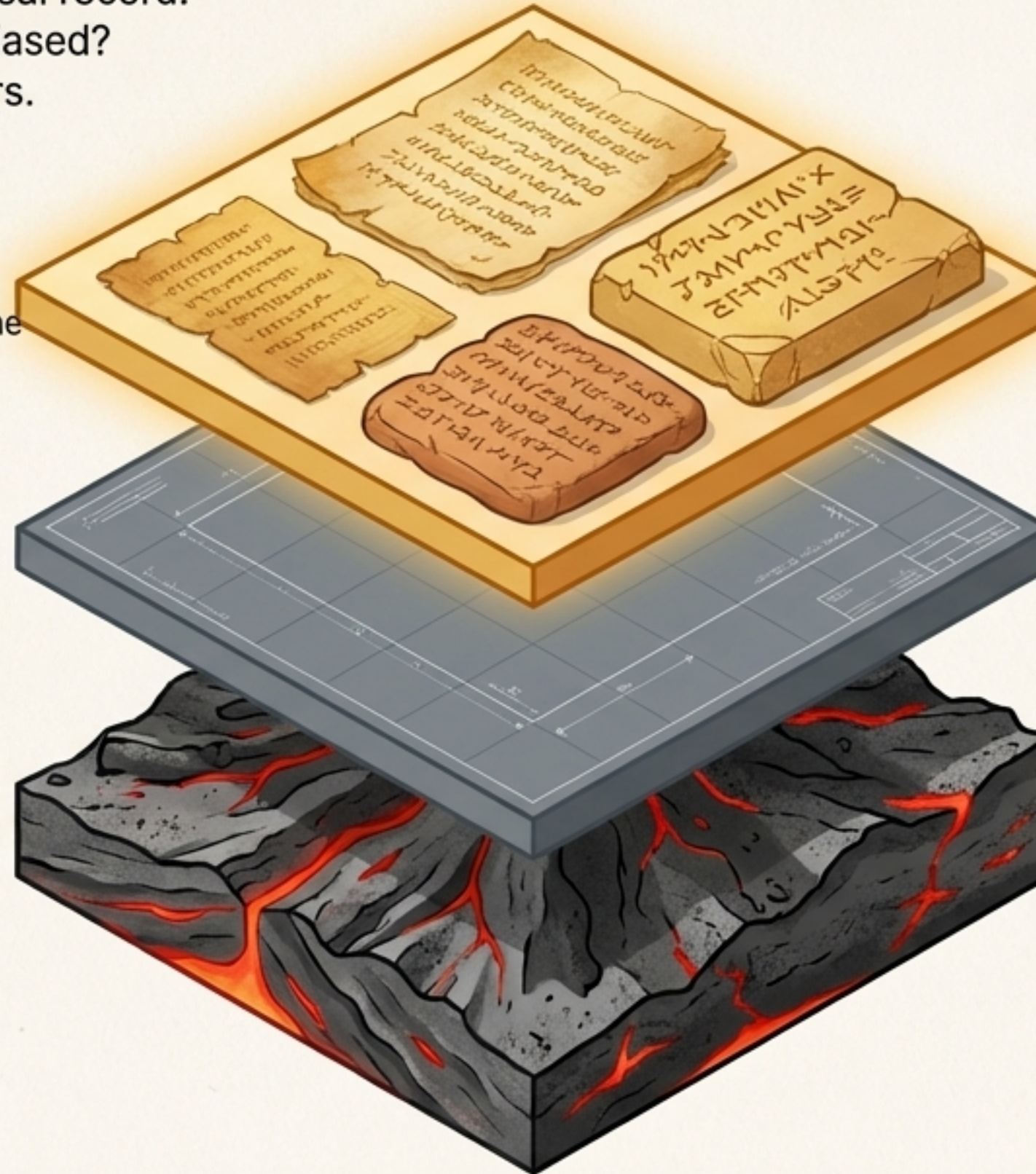
The specific types of text preserved in stone systematically exclude certain realities.
The focus of our computational research.

2. Survey Deficit

Institutional gaps. 100-200x fewer rescue excavations than comparable volcanic regions.

1. Volcanic Burial

Physical destruction hiding sites under ash at 2.4–6.2 mm/year.



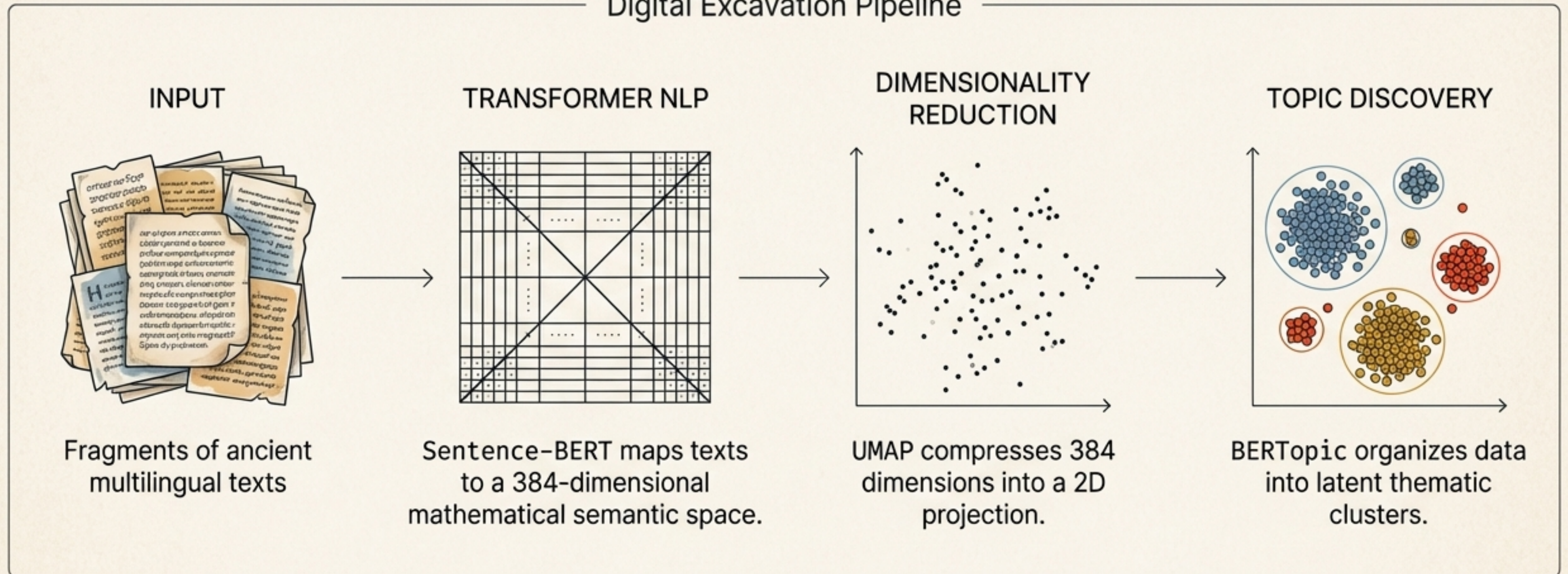
HYPOTHESIS:

If ancient civilizations are invisible in the dirt, can we find them hidden in the texts?

The Tool: Computational Textual Archaeology

Measuring shared concepts across 3,000 years, 12 cultures, and multiple languages without relying on exact word matches.

Digital Excavation Pipeline



The Two Worlds: A Tale of Two Corpora

The External Observers			The Indigenous Record	
1	Size	200 translated passages	Size	173 translated DHARMA inscriptions
2	Scope	12 traditions (Greek, Roman, Indian, Chinese, Arab, etc.)	Scope	Old Javanese, Sanskrit, Old Malay
3	Timespan	1700 BCE – 1365 CE	Timespan	Post-400 CE
4	Function	Geographic records, trade logs, maritime travel	Function	Royal decrees, tax exemptions, religious endowments

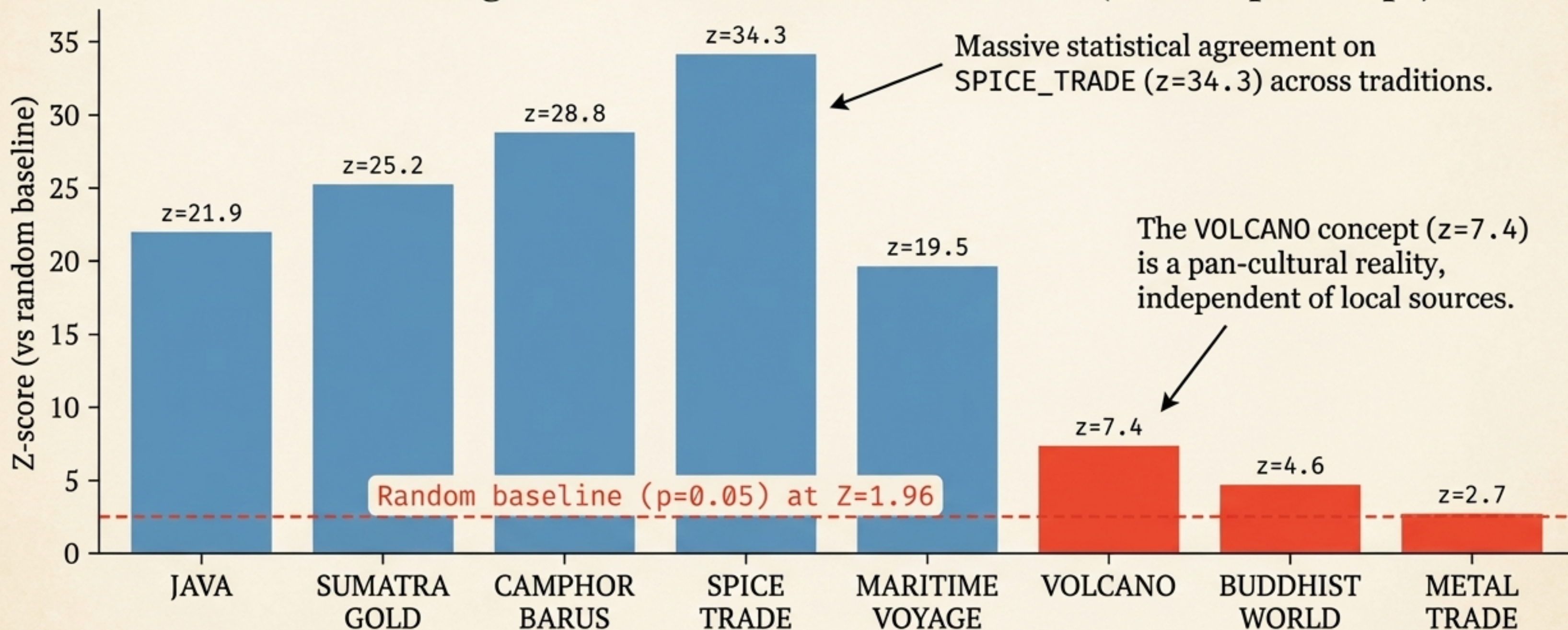
vs.

By embedding both corpora in the same semantic space, we can measure what external traditions converge on and what inscriptions systematically exclude.”

Finding 1: The Pan-Tradition Consensus

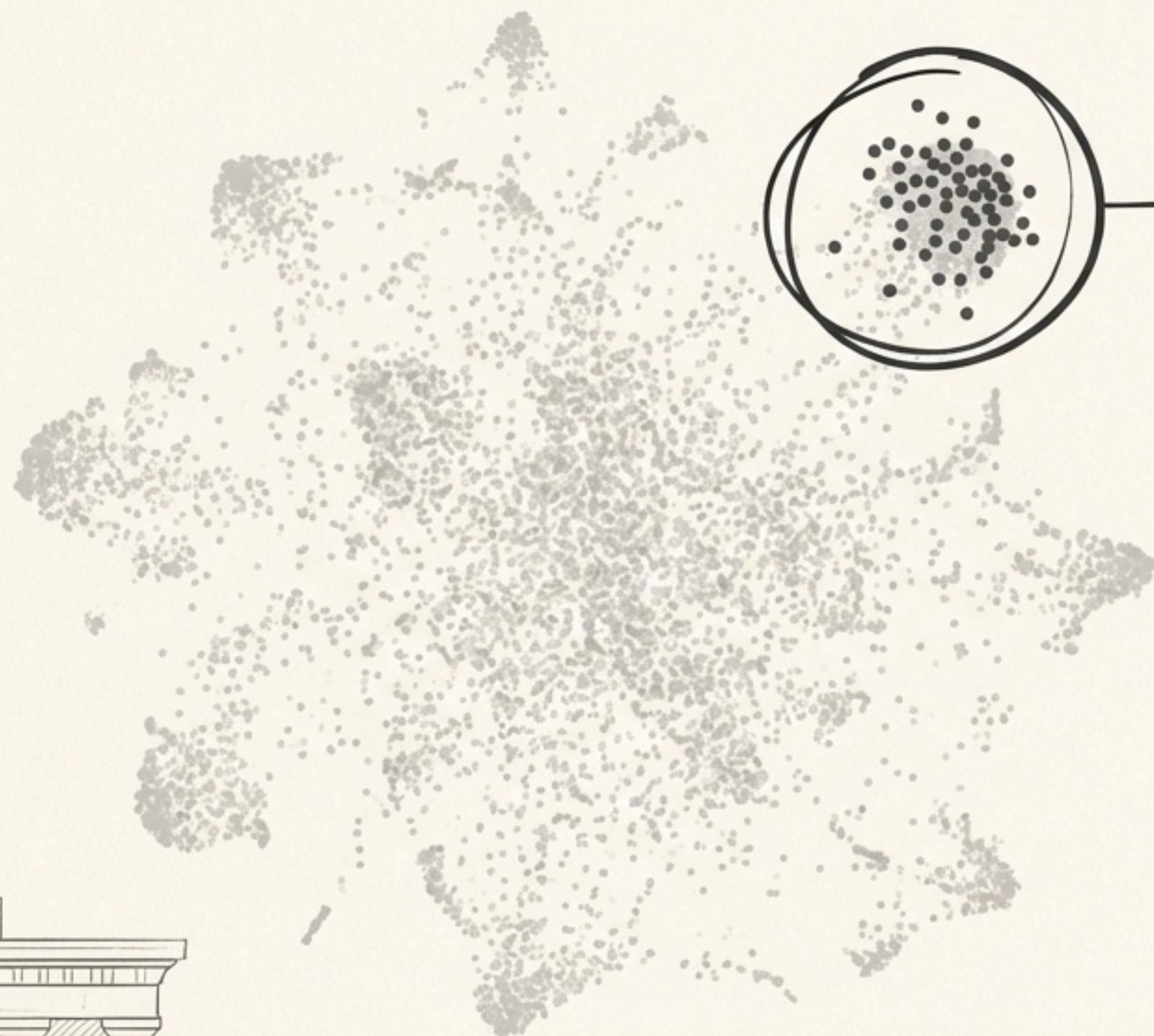
12 independent ancient traditions—from Greek geographers to Chinese dynastic historians—converged on a shared physical reality. They all saw the volcanoes.

Semantic Convergence Across 12 Ancient Traditions (8 Concept Groups)



Texts Cluster by Content, Not Culture

57% of all semantic clusters contain passages from multiple traditions. The algorithm groups texts by shared observation of the physical world, not by their origin language.

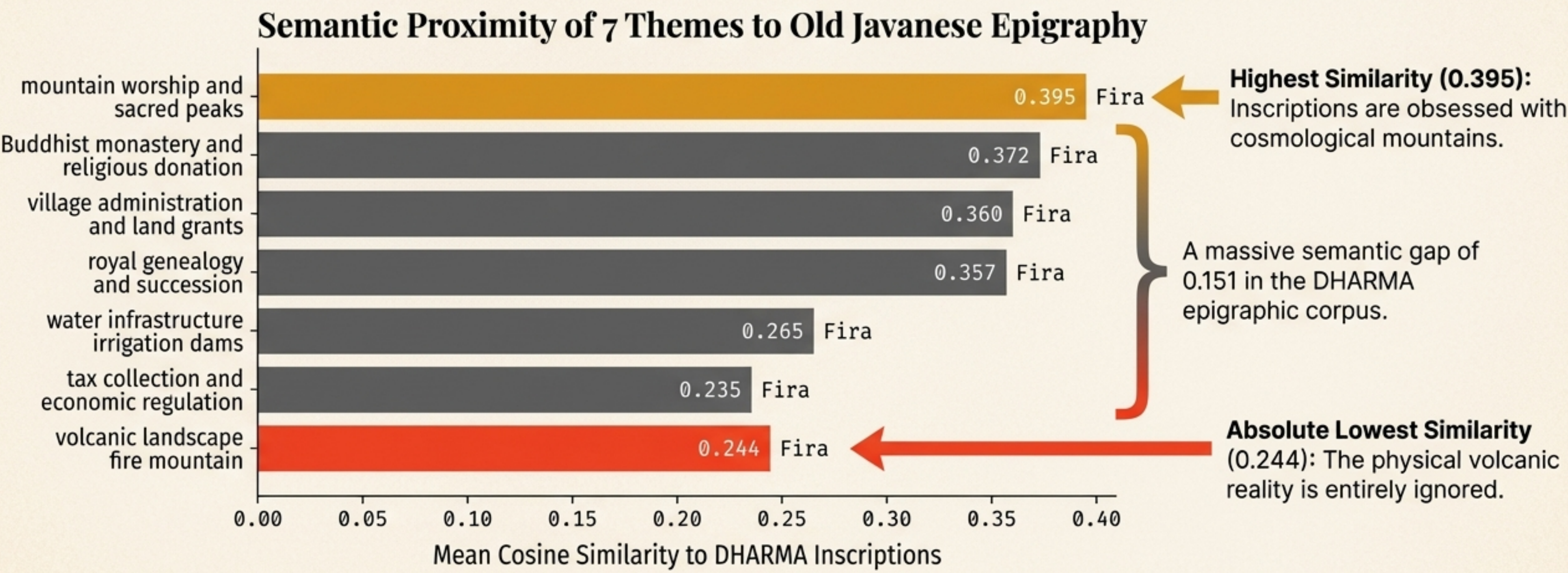


Topic 12: Mountain, slopes, clouds, smoke

- Contains texts from Greek, Arab, Chinese, and Indian sources.
- Demonstrates that diverse cultures intermingle perfectly when describing the same physical volcanic landscape.

Finding 2: The Deafening Silence in the Stone

Despite Java having 45 active volcanoes, the indigenous texts systematically exclude volcanic reality.

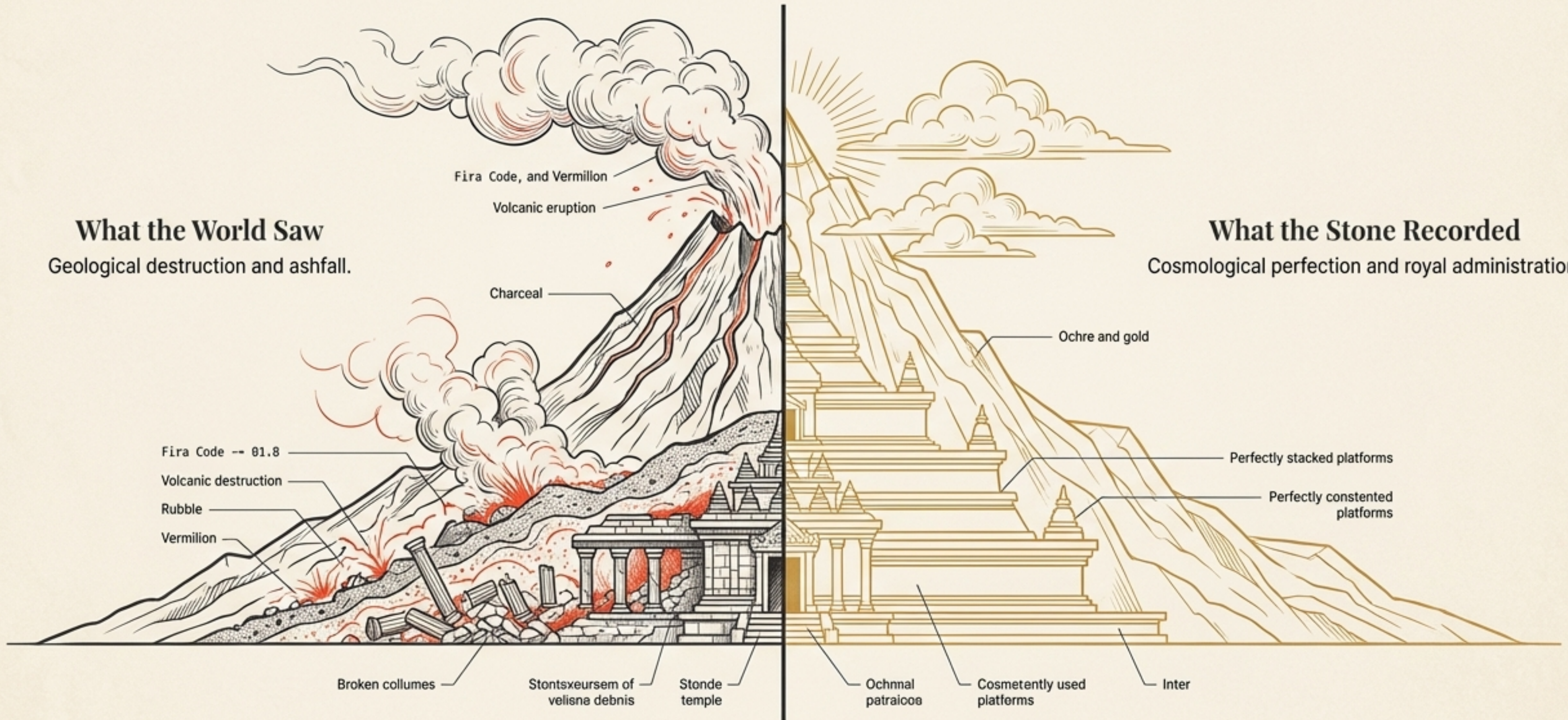


The Geologic Mountain vs. The Cosmological Peak

Ancient Javanese built temples directly on active volcanic slopes; they knew the volcanoes were there. The silence isn't ignorance. The genre of the stone inscription was designed strictly to record cosmological ideals, not geological realities.

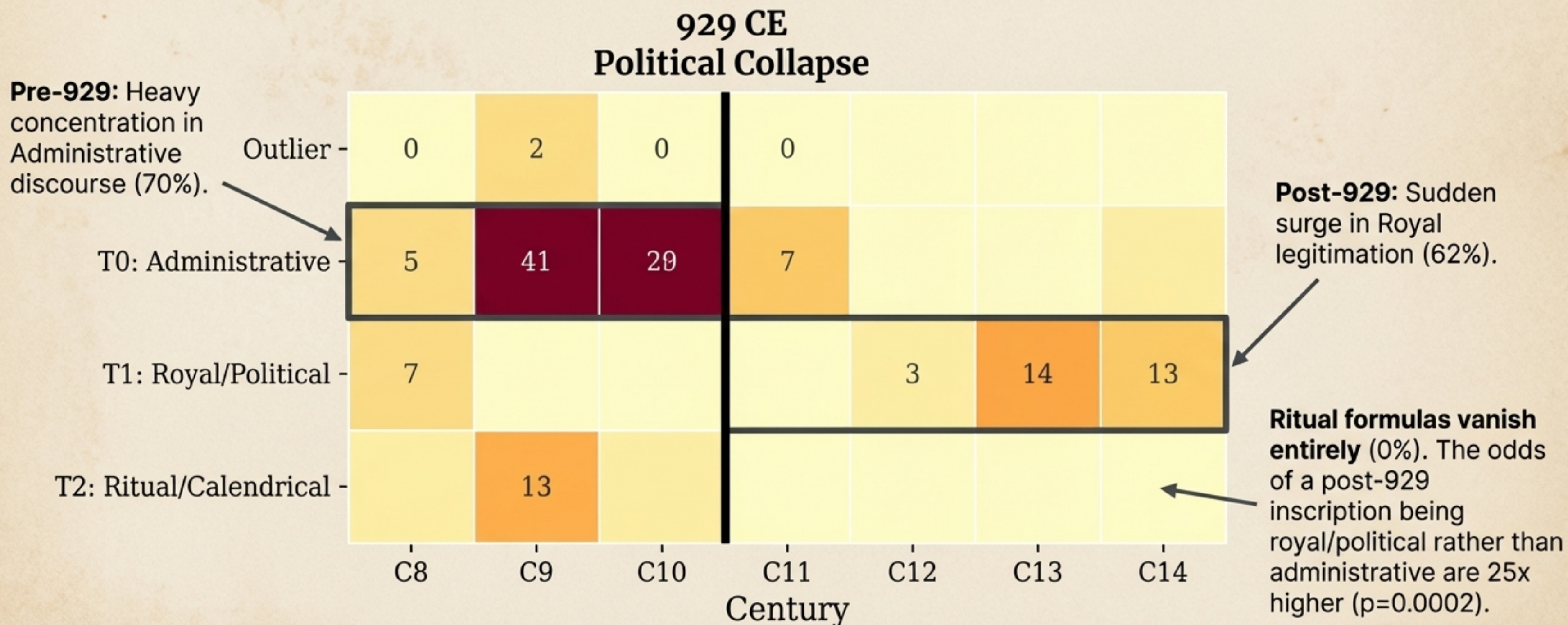
What the World Saw

Geological destruction and ashfall.



Finding 3: The 929 CE Mataram Collapse

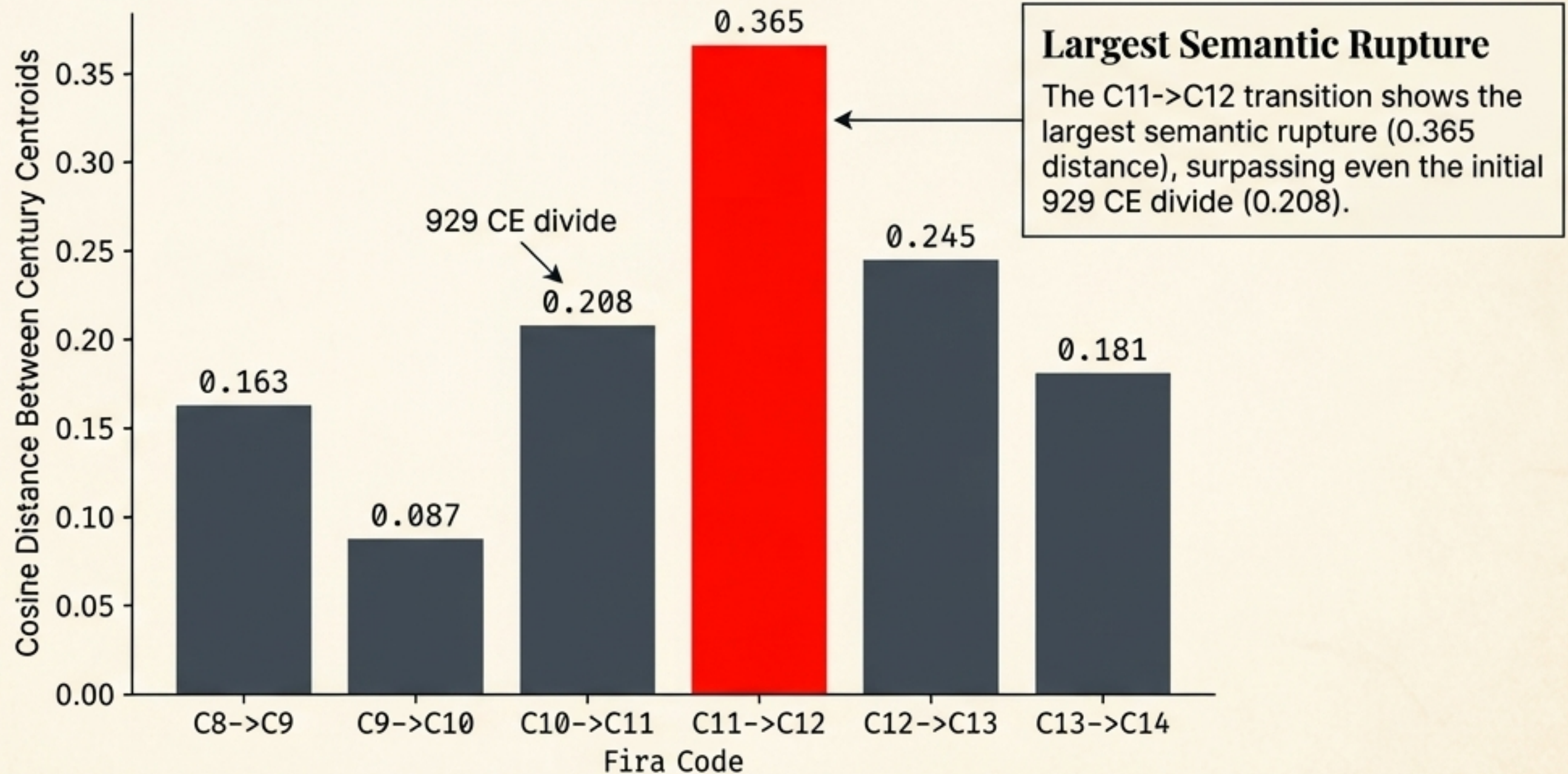
At the collapse of the kingdom, the subject matter etched into stone changed instantly. Political disruption fundamentally altered the genre rules of what an inscription was used for.



The Discursive Rupture

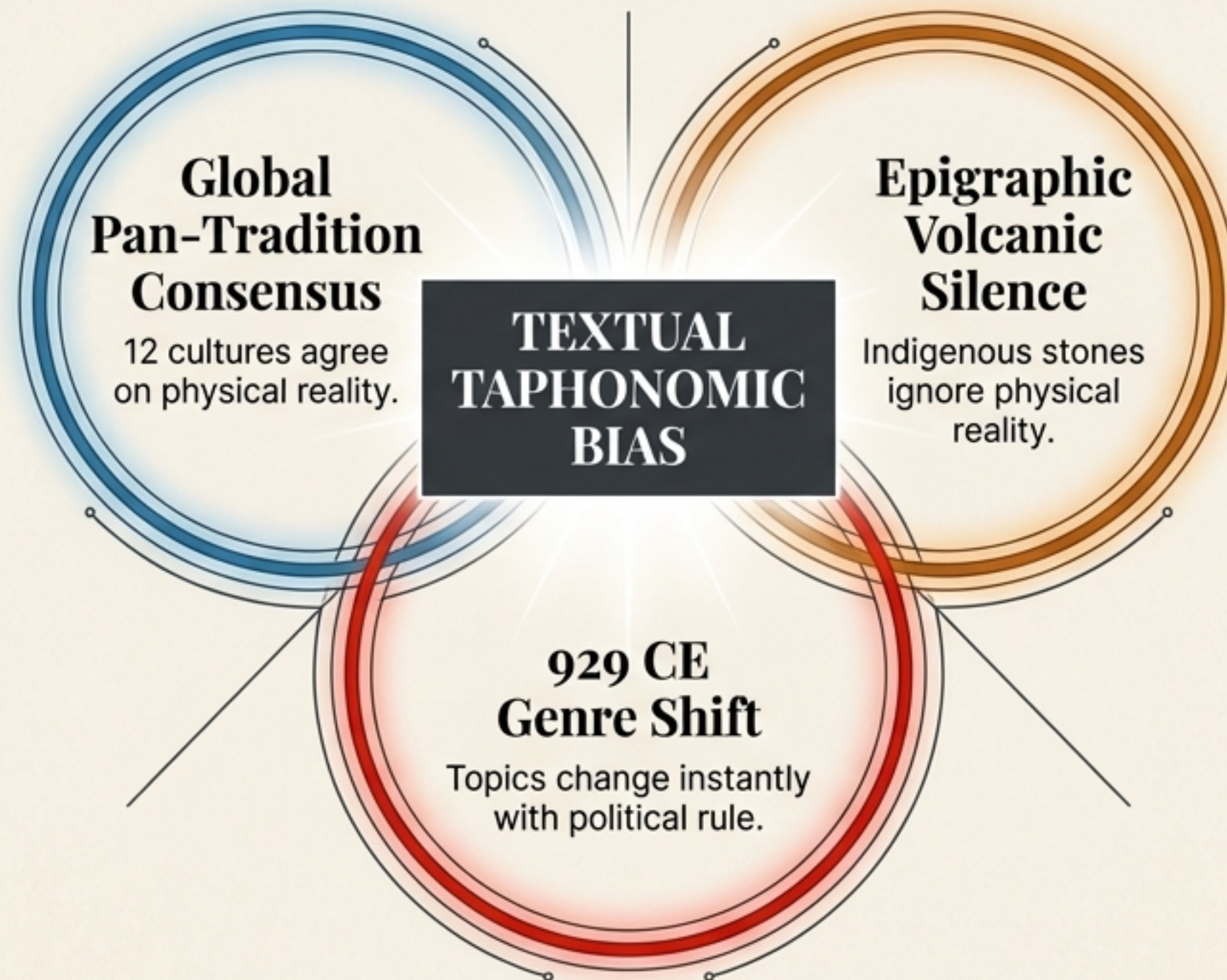
The transformation was a deliberate shift in discourse. The stone wasn't recording a changing physical environment; it was recording a changing political agenda.

Semantic Drift Across Centuries



The Stone Only Remembers Its Programming

Silence in the historical record is not evidence of absence. It is evidence of purpose. The Indonesian textual record is shaped by genre-specific selection pressures. Historical records operate like algorithms, filtering reality through strict rules of discourse.

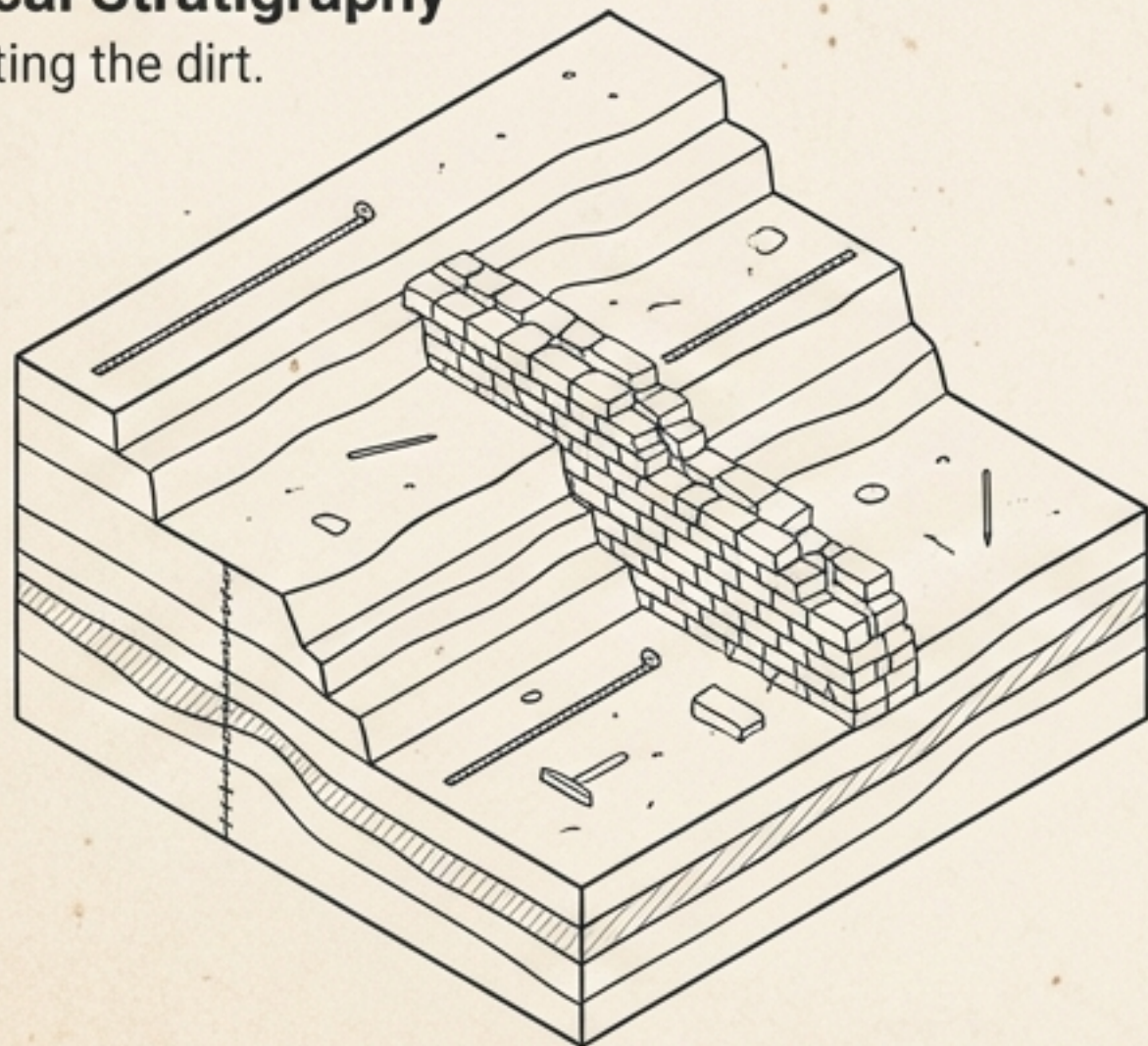


Why This Changes Archaeological Inference

We can no longer use inscription counts as direct proxies for population density or economic activity without accounting for genre bias. Comparing pre-929 CE administrative inscriptions to post-929 CE royal propaganda is comparing apples to oranges. The text is not the territory.

Physical Stratigraphy

Excavating the dirt.



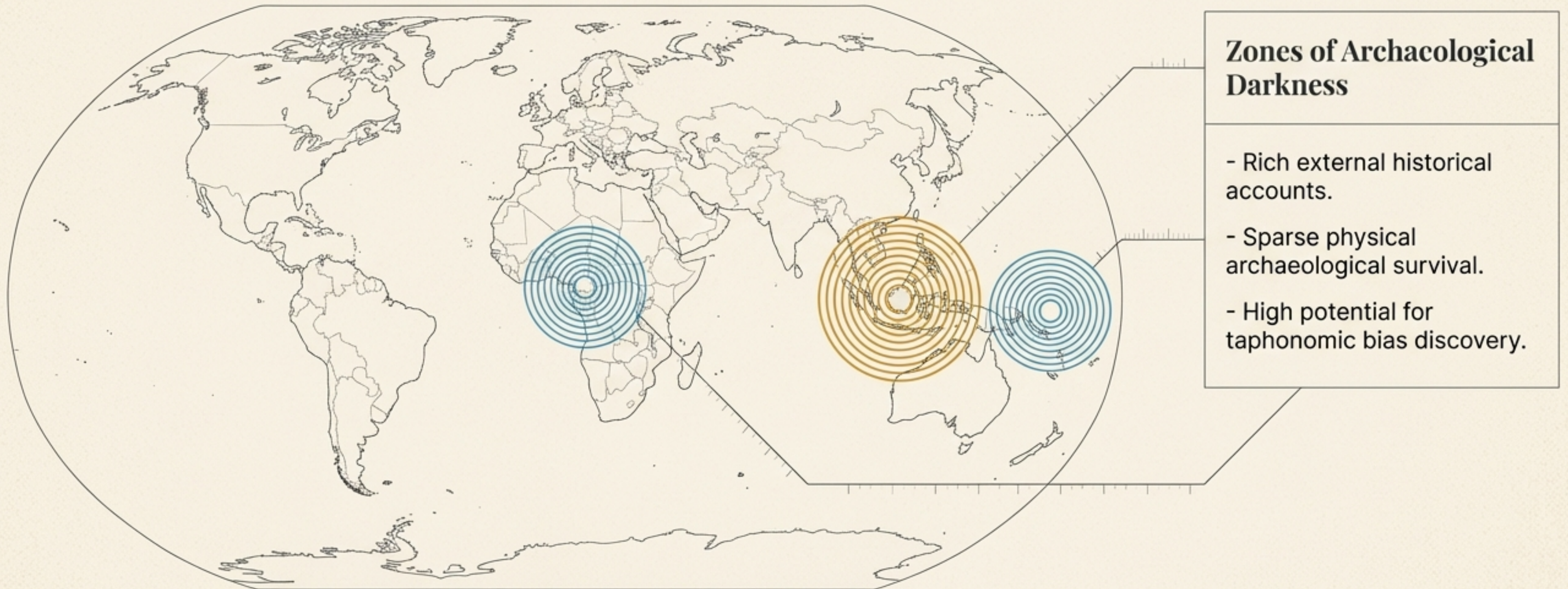
Digital Stratigraphy

Excavating the syntax.



A New Toolkit for Global History

The Computational Textual Archaeology pipeline (SBERT + UMAP + BERTopic) is fully reproducible. It opens a new frontier for digital humanities, allowing us to triangulate lost histories wherever a gap exists between external observation and internal silence.



Reproducibility & Open Data

Methodology Note

This research utilized an AI-augmented workflow, enabling a single researcher to execute 98 interdisciplinary experiments across archaeology, geology, and computational linguistics. It demonstrates the scale of pattern recognition now possible in the digital humanities.

The Open Data Links

VOLCARCH Repository

Cross-tradition corpus (E089 v5) and analysis scripts.

[🔗 github.com/mukhlisamien/volcarch](https://github.com/mukhlisamien/volcarch)

DHARMA EpiDoc Corpus

Digitized TEI-XML epigraphic inscriptions.

[🔗 github.com/erc-dharma](https://github.com/erc-dharma)